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# Removal of microcystin-LR cyanotoxin from water using magnesium oxide rice husk biochar composite

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## Abstract

The widespread occurrence of Microcystin-LR (MC-LR) in freshwater bodies has intensified concerns regarding its impact on drinking water quality and human health. This study developed a magnesium oxide (MgO)-modified rice husk biochar composite as an adsorbent for removing MC-LR from aqueous matrices. The composite was characterized using Scanning Electron Microscopy (SEM), Fourier Transform Infrared Spectroscopy (FTIR), and Brunauer–Emmett–Teller (BET) surface area analysis. A Central Composite Design (CCD) under Response Surface Methodology was employed in Design-Expert 13.0 to optimize operational parameters.

Characterization results confirmed that MgO modification enhanced the porous structure, yielding a hierarchical micro–mesoporous system with a specific surface area of 223.11 m<sup>2</sup>/g, indicating a substantial number of active adsorption sites. The highest removal efficiency (80.93%) was achieved at an initial MC-LR concentration of 25 ppb, an adsorbent dose of 20 mg, and a contact time of 60 min. The quadratic model demonstrated strong predictive capability ( $R^2 = 0.9559$ ), explaining 95.59% of the variability in removal efficiency. Adsorption performance was influenced by the interaction among concentration, adsorbent dosage, and contact time. Increased dosage enhanced removal due to greater availability of binding sites, whereas higher toxin concentrations led to site saturation and a plateau in efficiency. Overall, the MgO-modified biochar composite exhibited promising potential as an effective and sustainable adsorbent for MC-LR removal in water treatment applications.

**Keywords** Cyanobacteria, Cyanotoxin, Microcystin-LR, MgO biochar composite, Adsorption, Central composite design

## Introduction

Cyanobacteria, commonly associated with algal blooms, frequently proliferate in nutrient-rich aquatic environments. The increasing eutrophication of lakes and other aquatic systems is linked to the rising frequency, magnitude, and geographic distribution of cyanobacterial blooms worldwide. These blooms can greatly affect the ecosystem of the

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water bodies and can also pose serious threats to the environment and public health, particularly when toxin-producing species are involved [1].

Harmful Algal Blooms (HABs) are a specific and particularly problematic type of cyanobacterial proliferation that involves rapid, excessive growth of algae or cyanobacteria in freshwater and other aquatic systems. These blooms usually gather at or near the water surface and produce dense scums or discolor the water. The bloom refers to the rapid growth and subsequent decline of phytoplankton populations. HABs are considered harmful because of the negative effects on aquatic organisms, water quality, and the overall stability of the ecosystem. These often lead to lower dissolved oxygen concentrations, reduced water transparency, increased algal toxin production, and, in severe cases, mass mortality of aquatic organisms [2].

Moreover, the increase of toxic cyanobacterial blooms in drinking water supplies worldwide has reduced the safety of the water and increased concerns about health risks to marine organisms, animals, and humans [3]. On the other hand, cyanotoxins are a complex group of secondary metabolites produced by different genera of cyanobacteria that are highly toxic to a wide variety of species, including plants, animals, algae and humans [4]. These toxins can cause several health risks through drinking water, the food chain and recreational contact, such as possible damage to the kidneys, liver, and the central nervous system [5]. There are about 60% of cyanobacterial samples worldwide that contain cyanotoxins and are widely distributed in fresh and saltwater on different continents of the world, and microcystins are the most common cyanotoxin [6].

Furthermore, it has been documented that dissolved cyanotoxins are more resistant to traditional water treatment approaches, including filtration, sedimentation, and coagulation-flocculation [7]. Thus, cyanotoxins present a significant challenge for water treatment plants [8]. Among the methods used to remove cyanotoxins, adsorption has been considered an inexpensive, efficient, and straightforward approach for removing various pollutants from water without producing toxic intermediates/by-products [9]. The most often used technique for removing cyanotoxins, specifically microcystin, is adsorption [10]. Utilizing biochar as an adsorbent for cyanotoxin removal, such as microcystin, is a convenient and safe method, as biochar feedstocks are available, recyclable, and non-toxic [11]. Numerous modifications have been examined to increase biochar's remediation capacity and environmental advantages [12]. These changes include incorporating foreign materials into the biochar matrix to create a hybrid composite material. Then, these biochar composites are used to adsorb various contaminants [13]. Kar et al. (2025) successfully synthesized magnesium-modified rice husk biochar and reported its effective application for removing pesticides such as thiamethoxam, chlorpyrifos, and fenprothrin from aqueous solutions. Magnesium-modified biochar adsorption has some advantages over conventional water treatment technologies, including higher removal efficiencies, lower operating costs, easier application, and less secondary pollutant generation. Magnesium modification significantly improved the surface area, pore structure, and number of active adsorption sites in the biochar, thereby increasing its affinity for the target contaminants. These results highlight the great potential of magnesium-modified rice husk biochar as a sustainable, efficient, and environmentally friendly alternative to conventional water treatment techniques. Thus, its application to the removal of various emerging contaminants offers a promising approach for developing water purification technologies [14].

## Experimental

### Materials

Rice husk, a common agricultural byproduct, was sourced locally from the Tuburan Rice and Corn Mill in Villanueva, Misamis Oriental, Philippines. High-purity magnesium oxide (MgO, purity > 98%) was obtained from Shanghai Macklin Biochemical Technology Co., Ltd., China. The chemical was supplied as a powder and stored in airtight containers at room temperature to maintain its integrity and prevent moisture uptake. Microcystin-LR (MCLR, purity > 95%) was purchased from Gold Standard Diagnostics, Spain.

### Synthesis and characterization of magnesium oxide-rice husk biochar composite (MgO-BCR)

The raw rice husk was thoroughly washed with distilled water to remove dust and other impurities. The rice husk was first air-dried overnight at ambient temperature to remove surface moisture. Subsequently, it was oven-dried at 105 °C until a constant weight was achieved, ensuring the complete removal of residual moisture. The dried rice husk was pulverized and passed through an 80-mesh sieve to obtain a uniform particle size. The sieved material was then subjected to pyrolysis at 500 °C for 2 h in a muffle furnace. After pyrolysis, the resulting biochar was allowed to cool naturally to room temperature. The biochar was subsequently rinsed three times with deionized (DI) water to remove any residual impurities. Finally, it was oven-dried at 105 °C until a constant weight was achieved.

To modify the biochar, 0.6 g of magnesium oxide was dissolved in 200 mL of deionized (DI) water. The suspension was magnetically stirred for 10 to 15 min at room temperature to ensure complete dispersion of MgO particles. Subsequently, 20 g of previously prepared rice husk biochar was added to the MgO suspension. The resulting mixture was subjected to ultrasonic treatment for 30 min and was further magnetically stirred for 1 h at room temperature. The mixture was then left to settle for 24 h. The mixture was then oven-dried at 105 °C until a constant weight was achieved, then pyrolyzed in a muffle furnace at 500 °C for 1 h. After pyrolysis, the MgO-BCR was allowed to cool naturally to room temperature before further use or characterization.

The MgO biochar composite was characterized using analytical techniques to comprehensively evaluate its properties. The Brunauer-Emmett-Teller (BET) method was employed to determine the specific surface area and porosity of the material. Morphological analysis was conducted using scanning electron microscopy (SEM) to observe the surface structure and particle distribution. Additionally, Fourier transform infrared spectroscopy (FTIR) was used to identify characteristic functional groups on the surface of MgO-BCR, providing insight into its chemical composition and surface chemistry.

### Adsorption experiments

A stock solution of microcystin-LR (MCLR) was prepared at 1,000 µg/L by diluting the standard solution with deionized water. Central Composite Design (CCD) was employed as an optimization tool in RSM using Design-Expert software 13.0. Initial MCLR concentrations ranged from 25 to 65 µg/L and were prepared by diluting the stock solution with deionized water. The adsorbent dose was varied between 10 and 30 mg, and the contact time was set between 40 and 80 min. Adsorption experiments were conducted

**Table 1** Experimental range and levels of independent variables in adsorption

| Factor                                   | Coded Level |    |    |    |    |
|------------------------------------------|-------------|----|----|----|----|
|                                          | -2          | -1 | 0  | 1  | 2  |
| Initial microcystin concentration (µg/L) | 25          | 35 | 45 | 55 | 65 |
| Adsorbent dose (mg)                      | 10          | 15 | 20 | 25 | 30 |
| Contact time (min)                       | 40          | 50 | 60 | 70 | 80 |

**Table 2** Experimental runs for microcystin-LR removal using magnesium oxide biochar composite

| Run | Operating Variables                        |                     |                    |
|-----|--------------------------------------------|---------------------|--------------------|
|     | Initial Microcystin-LR Concentration (ppb) | Adsorbent Dose (mg) | Contact Time (min) |
| 1   | 45                                         | 20                  | 80                 |
| 2   | 45                                         | 20                  | 60                 |
| 3   | 25                                         | 20                  | 60                 |
| 4   | 55                                         | 15                  | 50                 |
| 5   | 45                                         | 30                  | 60                 |
| 6   | 65                                         | 20                  | 60                 |
| 7   | 35                                         | 15                  | 50                 |
| 8   | 35                                         | 15                  | 70                 |
| 9   | 35                                         | 25                  | 50                 |
| 10  | 35                                         | 25                  | 70                 |
| 11  | 45                                         | 20                  | 40                 |
| 12  | 55                                         | 25                  | 50                 |
| 13  | 55                                         | 25                  | 70                 |
| 14  | 55                                         | 15                  | 70                 |
| 15  | 45                                         | 20                  | 60                 |
| 16  | 45                                         | 20                  | 60                 |
| 17  | 45                                         | 10                  | 60                 |

in a beaker containing 20 ml of the required initial MCLR concentrations and varying adsorbent doses, as specified in the CCD matrix. All beakers were covered with aluminum foil during the experiments. After the contact time, suspensions were filtered through 0.22 µm pore-size polyethersulfone (PES) syringe filters, and the filtrates were collected for MCLR analysis. The experimental range and levels are shown in Table 1. The removal efficiency (% removal) was calculated using the following equation:

$$\% \text{ removal} = \frac{C_o - C_e}{C_o} \times 100 \quad (1)$$

where  $C_o$  is the initial concentration (ug/L), and  $C_e$  is the equilibrium concentration (ug/L).

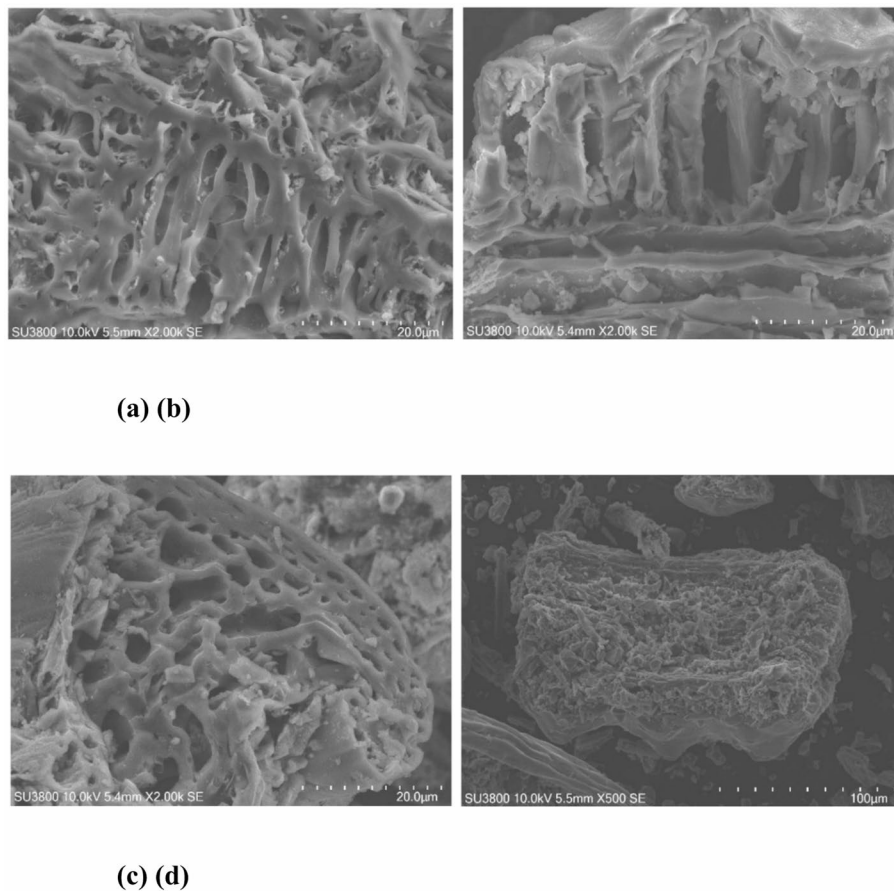
This study aimed to determine the removal efficiency of Microcystin-LR using magnesium oxide biochar composite from rice husks. Using the values of the coded level of independent variables in Table 1, seventeen (17) experimental runs were generated and conducted sequentially in the laboratory (Table 2).

## Results and discussion

### Physical and chemical properties of MgO-biochar composite

#### Porosity

The surface morphology of rice husk (RH) biochar is visible in SEM images, which show a network of interconnected pores at all magnification levels. The images at 2000× with a 20 µm scale bar (Fig. 1a and b, and 1c) reveal a sponge-like structure and a honeycomb



**Fig. 1** SEM images RH Biochar composites at (a, b, and c) 20 μm and (d) 100 μm

pattern of pores, which contain both long, narrow channels and circular spaces [15]. The network of interconnected pores exhibits complex internal porosity, providing multiple sites for adsorption. The 500× magnification at 100 μm scale bar (Fig. 1d) provides an expanded view, revealing both large voids and spaces between particles, which demonstrates that porosity extends throughout the entire structure [16].

MgO modification effectively enhanced the biochar's porous structure, confirming its role as a functional enhancer. The biochar exhibits a hierarchical structure of micro to macropores, indicating that a multi-scale porous system exists for adsorption and mass transfer activities [15].

#### **Particle shape**

The SEM images (Fig. 1) show that RH biochar particles exhibit irregular, heterogeneous surface patterns. The particles at higher magnification display slab-like, rectangular shapes, characterized by rough surfaces with cracks, jagged edges, and fragmented layers. The biochar appears as a large, clumped mass at 500× magnification and a 100 μm scale, revealing layered structures that indicate some degree of pyrolysis-induced collapse [16]. The surface exhibits a rough texture, characterized by multiple deep cracks and sharp edges, resulting in an uneven surface structure. The combination of slab-like structures and broken particle surfaces creates additional surface area that works in conjunction with the internal pore system visible at higher magnification. The material

exhibits strong adsorption capabilities due to its structure, which combines porous channels with irregularly shaped particles [15].

### Functional group

The identification of surface functional groups on the RH Biochar Composite was conducted using FTIR spectroscopy over the wavenumber range of 4000–400  $\text{cm}^{-1}$  under dry conditions, as illustrated in Fig. 2. The spectrum exhibited two main absorption bands, appearing at 793.84  $\text{cm}^{-1}$  and 1050.34  $\text{cm}^{-1}$ .

The peak at 793.84  $\text{cm}^{-1}$  indicates aromatic C–H out-of-plane bending, which shows condensed aromatic structures exist in thermally treated biomass [17]. The more intense peak at 1050.34  $\text{cm}^{-1}$  corresponds to C–O stretching vibrations, indicative of alcohols, phenols, or carboxylic acid derivatives, which are often retained or formed during pyrolysis [18].

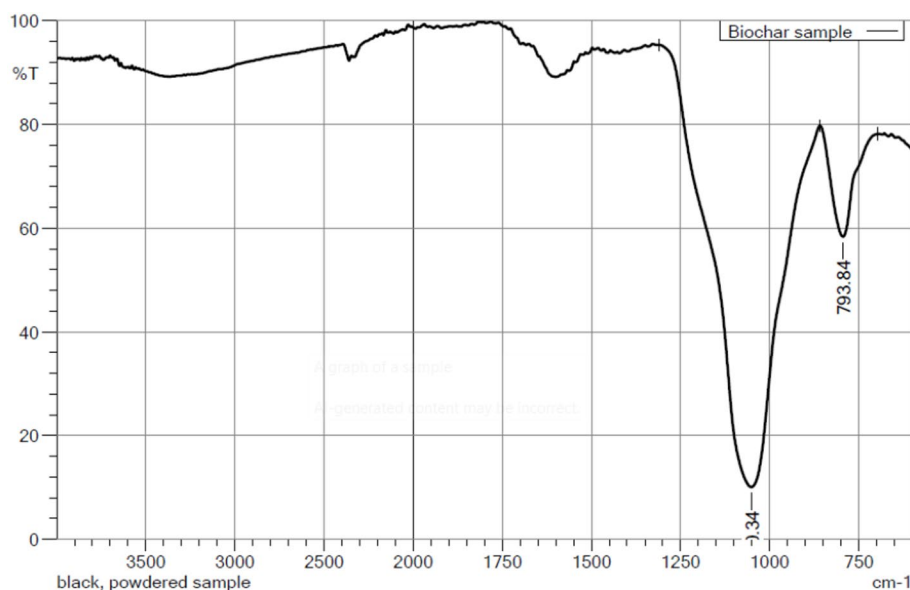
The functional groups on biochar surfaces enable them to react with substances, creating strong bonds that help remove contaminants and improve water quality.

Peaks at 793.84  $\text{cm}^{-1}$  and 1050.34  $\text{cm}^{-1}$ .

The FTIR spectral analysis of RH Biochar Composite reveals surface functional groups, including aromatic and oxygenated components, that confer strong adsorption capacity towards Microcystin-LR. The composite structure comprises various groups that generate active bonding sites, enabling hydrogen bonds, electrostatic interactions, and  $\pi$ – $\pi$  stacking to remove cyanotoxins from water [18, 19].

### Surface area

The BET analysis indicates that the biochar sample possesses a specific surface area of 223.11  $\text{m}^2/\text{g}$ , reflecting a well-developed porous structure. This surface area is within the range commonly reported for engineered or activated biochar, suggesting a substantial availability of surface sites for adsorption and interfacial reactions. Such a high surface area is advantageous for applications involving adsorption, soil amendment, catalysis



**Fig. 2** Fourier Transform Infrared (FTIR) Spectrum of RH Biochar Composite

support, and contaminant immobilization, as it enhances interaction between the biochar surface and surrounding species.

**Pore volume and average pore diameter**

The total pore volume, measured at 0.127 cm<sup>3</sup>/g, further supports the presence of a porous network within the biochar matrix. This pore volume indicates moderate pore development, which contributes to mass transfer and accessibility of internal surfaces. While the pore volume is not extremely high, it is sufficient to complement the relatively large surface area, implying that the pores are finely distributed rather than dominated by large macropores.

The average pore diameter of 2.2764 nm places the biochar predominantly in the mesoporous range (2–50 nm) according to the IUPAC classification. Mesoporosity is particularly beneficial for adsorption processes involving larger molecules and facilitates improved diffusion compared to purely microporous materials. The presence of mesopores also suggests that the biochar structure may result from devolatilization and partial pore widening during pyrolysis.

Overall, the combination of high BET surface area, moderate pore volume, and mesoporous structure indicates that the biochar exhibits favorable textural properties for environmental and agricultural applications. These characteristics enhance its capacity to adsorb nutrients or pollutants and improve its functional performance when used as a soil conditioner or adsorbent. The BET results, therefore, confirm that the produced biochar has a well-developed porous architecture suitable for practical utilization.

**Percent removal of microcystin-LR using magnesium-oxide biochar composite**

The experimentation was conducted using the CCD design shown in Table 2. The percentage removal of Microcystin-LR from aqueous solution ranged from 16.68% to 80.93% (Tables 3 and 4). The interactive effects of the operating variables influenced the increase and decrease in Microcystin-LR removal. A previous study also reported similar findings, highlighting how combined operating factors affect Microcystin-LR removal in the water. These results confirm that the selected variables play a significant role in the efficiency of Microcystin-LR removal using Magnesium-Oxide Biochar Composite from rice husk.

The maximum removal efficiency of 80.93% can be considered moderate compared to highly engineered adsorbents, but it is important to put this result in the context of the material’s nature. Previous studies reported that granular activated carbon (GAC) demonstrated high Microcystin-LR adsorption performance due to its mesoporous structure and optimized pore distribution [20], while graphene nanoplatelets achieved approximately 99% removal efficiency and exhibited adsorption capacities nearly 100 times greater than conventional GAC [21]. Despite the superior performance of these advanced materials, the MgO-modified rice husk biochar composite developed in this study offers practical advantages including the utilization of abundant agricultural

**Table 3** FTIR functional groups of RH biochar composite

| Wavelength Range (cm <sup>-1</sup> ) | Functional Groups      | Band                 | Interpretation                             |
|--------------------------------------|------------------------|----------------------|--------------------------------------------|
| 793.84                               | Aromatic C-H           | Out-of-plane bending | Indicates aromatic ring structures         |
| 1050.34                              | C-O (alcohols/phenols) | Stretching           | Suggest oxygenated surface functionalities |

**Table 4** Percent removal of microcystin-LR using magnesium oxide biochar composite

| Run | Operating Variables                        |                     |                    | % Removal |
|-----|--------------------------------------------|---------------------|--------------------|-----------|
|     | Initial Microcystin-LR Concentration (ppb) | Adsorbent Dose (mg) | Contact Time (min) |           |
| 1   | 45                                         | 20                  | 80                 | 24.98     |
| 2   | 45                                         | 20                  | 60                 | 48.25     |
| 3   | 25                                         | 20                  | 60                 | 80.93     |
| 4   | 55                                         | 15                  | 50                 | 77.06     |
| 5   | 45                                         | 30                  | 60                 | 74.04     |
| 6   | 65                                         | 20                  | 60                 | 25.46     |
| 7   | 35                                         | 15                  | 50                 | 27.19     |
| 8   | 35                                         | 15                  | 70                 | 67.50     |
| 9   | 35                                         | 25                  | 50                 | 29.20     |
| 10  | 35                                         | 25                  | 70                 | 16.68     |
| 11  | 45                                         | 20                  | 40                 | 56.61     |
| 12  | 55                                         | 25                  | 50                 | 63.62     |
| 13  | 55                                         | 25                  | 70                 | 72.18     |
| 14  | 55                                         | 15                  | 70                 | 54.07     |
| 15  | 45                                         | 20                  | 60                 | 41.27     |
| 16  | 45                                         | 20                  | 60                 | 56.15     |
| 17  | 45                                         | 10                  | 60                 | 60.41     |

waste, lower production cost, simpler synthesis process, and improved environmental sustainability, making it more suitable for scalable and cost-effective water treatment applications.

Moreover, in the study of Yuan et al. (2021), HCl-treated bamboo biochar (HBB) is one of the biochars tested for sorption and desorption, which is significantly less effective in the removal of microcystin-LR in comparison with other materials, with removal efficiencies as low as 22.6%. The poor performance is due to the limited mesoporosity and less favorable point of zero charge, which hinder the electrostatic interactions and pore-filling ability for the toxin capture. Moreover, although some biochars exhibit a high resistance to the release of the sorbed toxin, the specific structural features of the bamboo-based biochar lead to a less effective trapping ability, which is not suitable for high-efficiency water treatment applications [11]. In contrast, the MgO–rice husk biochar developed in the present study achieved a substantially higher removal efficiency of 80.93%, suggesting that the incorporation of MgO and the utilization of rice husk as a precursor significantly enhanced the adsorption performance through improved surface properties, pore structure, and stronger interactions with microcystin-LR.

#### Model fitting on the microcystin-LR removal using magnesium oxide biochar composite

The Central Composite Design (CCD) indicated that a quadratic model best fits the prediction of Microcystin-LR removal via adsorption. This is supported by the statistical analysis presented in Table 5, which uses ANOVA.

The  $R^2$  value of 0.9559 indicates high accuracy in predicting percent removal, suggesting that the model reliably captures the data. This implies that the model can explain 95.59% of the variability in the response.

The quadratic surface model equation (Eq. 1) presented below is derived from the actual factor levels generated with Design-Expert 13.0.

**Table 5** ANOVA results for evaluating the fit of the quadratic response surface model

| Source                  | Sum of Squares | df | Mean Square | F Value | P-Value Prob > F    |
|-------------------------|----------------|----|-------------|---------|---------------------|
| Model                   | 4737.14        | 9  | 526.35      | 9.63    | 0.0217 <sup>a</sup> |
| A-Initial Concentration | 528.29         | 1  | 528.29      | 9.66    | 0.0359              |
| B-Adsorbent Dose        | 1255.67        | 1  | 1255.67     | 22.96   | 0.0087              |
| C-Contact Time          | 707.69         | 1  | 707.69      | 12.94   | 0.0228              |
| AB                      | 1310.41        | 1  | 1310.41     | 23.97   | 0.0081              |
| AC                      | 49.86          | 1  | 49.86       | 0.9119  | 0.3937              |
| BC                      | 180.59         | 1  | 180.59      | 3.30    | 0.1433              |
| A <sup>2</sup>          | 1630.78        | 1  | 1630.78     | 29.82   | 0.0055              |
| B <sup>2</sup>          | 88.98          | 1  | 88.98       | 1.63    | 0.2711              |
| C <sup>2</sup>          | 63.21          | 1  | 63.21       | 1.16    | 0.3428              |
| Residual                | 218.72         | 4  | 54.68       |         |                     |
| Lack of Fit             | 107.87         | 2  | 53.93       | 0.9731  | 0.5068 <sup>b</sup> |
| Pure Error              | 110.85         | 2  | 55.43       |         |                     |
| Cor Total               | 4955.86        | 13 |             |         |                     |
| R <sup>2</sup> = 0.9559 |                |    |             |         |                     |

*a* = significant; *b* = not significant

$$y = 731.07196 - 18.36335A - 17.79744B - 2.05777C + 0.297218AB + 0.028988AC + 0.110337BC + 0.127980A^2 - 0.119575B^2 - 0.018117C^2 \quad (2)$$

where  $y$  represents the predicted percent removal of Microcystin-LR in an aqueous solution, at the same time,  $A$ ,  $B$ , and  $C$  correspond to the initial Microcystin-LR concentration (ppb), adsorbent dose (mg), and contact time (min), respectively.

This model accounts for the main effects of each factor ( $A$ ,  $B$ ,  $C$ ), their quadratic effects ( $A^2$ ,  $B^2$ ,  $C^2$ ), and the pairwise interactions between factors ( $AB$ ,  $AC$ ,  $BC$ ). The coefficients indicate the magnitude and direction of each term's influence on percent removal.

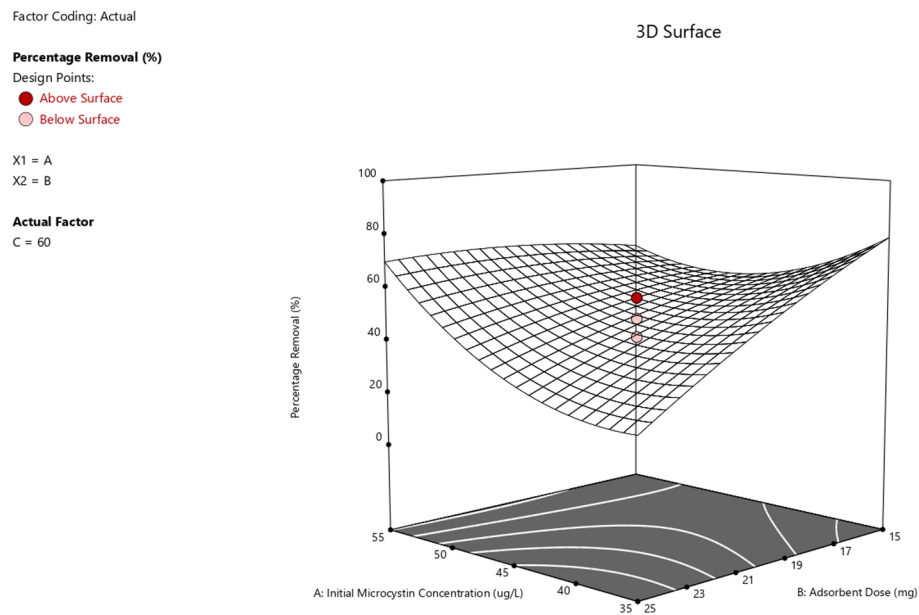
Negative coefficients for  $A$ ,  $B$ , and  $C$  suggest that increasing these parameters individually leads to a decrease in the predicted removal, while positive interaction terms –  $AB$ ,  $AC$ , and  $BC$  indicate synergistic effects between the corresponding variables. The quadratic terms account for nonlinear effects, allowing the model to capture curvature in the response surface.

Overall, Eq. 1 enables prediction and optimization of Microcystin-LR removal under different experimental conditions and can be used in conjunction with response surface methodology for process optimization.

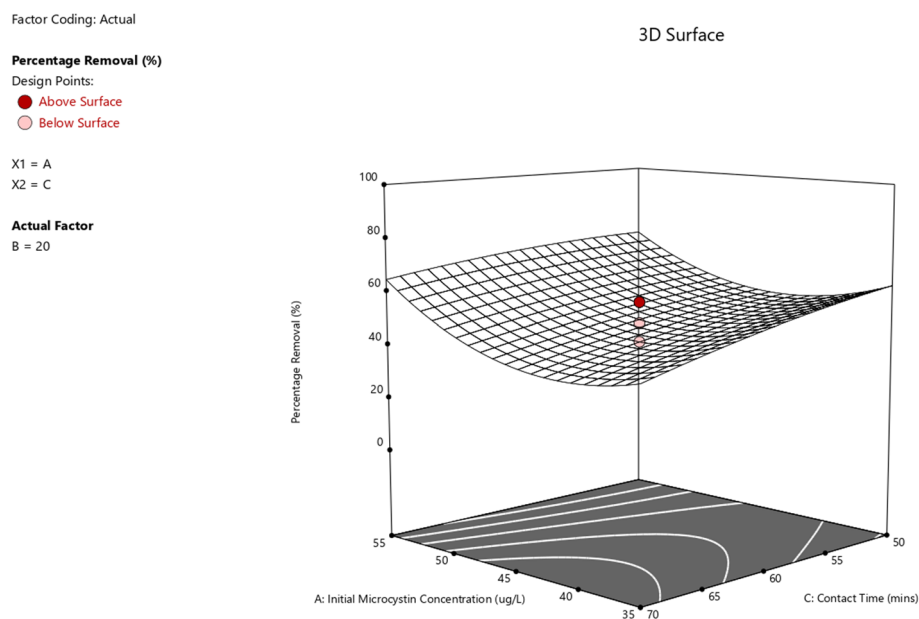
#### Interactive effects of initial microcystin-LR concentration and adsorbent dose

The effect of initial concentration and adsorbent dose on the removal of Microcystin-LR using Magnesium Oxide Biochar Composite is presented in Fig. 3.

Based on the graph shown, the adsorption efficiency is affected by both the amount of adsorbent and the Initial microcystin-LR concentration. The higher the amount of biochar used, the more binding sites are available, hence better removal efficiency. However, with increased concentrations of toxins in the solution, the efficiency plateaus due to the saturation of binding sites [22]. The response surface plot indicates that a combination of the adsorbent amount and the toxin concentration is necessary for optimal removal.



**Fig. 3** 3D response surface interactive effect of initial microcystin-LR concentration and adsorbent dose

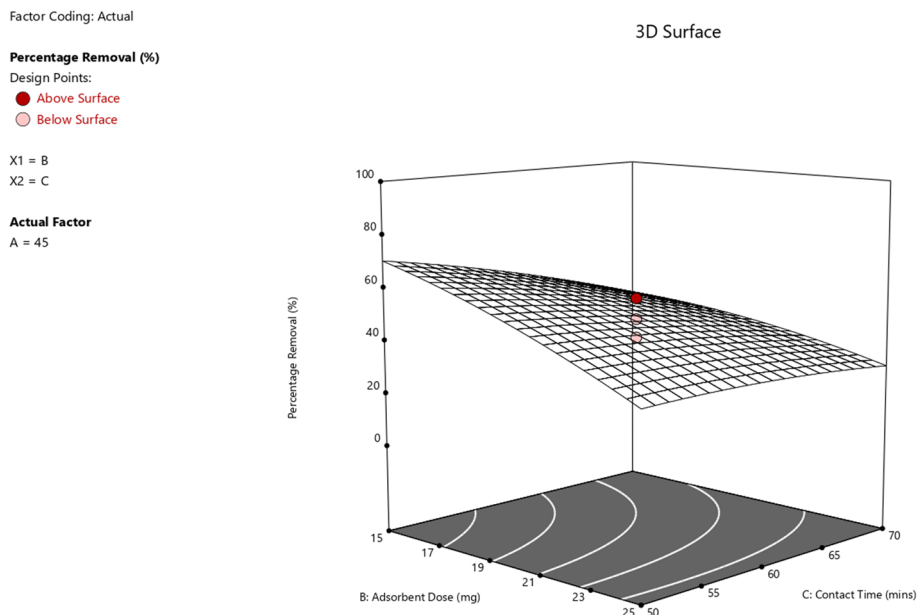


**Fig. 4** 3D Response Surface Interactive Effect of Initial Microcystin-LR Concentration and Contact Time

#### Interactive effects of initial microcystin-LR concentration and contact time

The effect of initial concentration and contact time on the removal of Microcystin-LR by Magnesium-Oxide Biochar Composite was presented in Fig. 4.

Based on the graph shown, adsorption efficiency depends on both the starting concentration of Microcystin-LR and the contact time. At lower concentrations, giving the system more time allows the biochar to remove most of the toxin. As the concentration rises, the percentage removal drops and then levels off, showing that the adsorbent's binding sites are gradually filled [19]. The graph makes it clear that longer contact times



**Fig. 5** 3D response surface on the effect of contact time and adsorbent dose

help, but they cannot fully overcome the limits imposed by high toxin concentration loads.

#### Interactive effects of adsorbent dose and contact time

The effect of adsorbent dose and contact time on the removal of Microcystin-LR by Magnesium-Oxide Biochar Composite was presented in Fig. 5.

Based on the graph shown, adsorption efficiency depends on both the amount of biochar and the contact time. Adding more adsorbent provides more sites for the toxin to bind, improving removal when sufficient time is allowed. At short contact times, even higher doses cannot reach full efficiency because the process has not yet settled [19]. Longer contact times increase removal, but the improvement slows once the binding sites are filled [23]. The graph shows that the best results come from balancing dose and time, so the adsorbent has both the capacity and the opportunity to capture Microcystin-LR.

#### Conclusions

The MgO-modified rice husk biochar showed great potential for the adsorption of Microcystin-LR from aqueous solutions. Surface characterization showed that the addition of MgO increased surface area and porosity, providing many adsorption-active sites. It was found that the adsorbent dose, initial toxin concentration, and contact time had significant interaction effects, and that the maximum removal efficiency of 80.93% was achieved through optimization using Response Surface Methodology. The developed quadratic model was highly predictive ( $R^2=0.9559$ ) and confirmed the experimental results. Overall, the results indicate that the MgO-biochar composite is a sustainable and efficient material for removing MC-LR, with potential applications in water treatment systems to reduce cyanotoxin contamination. In this study, the main focus is on the application of MgO-biochar from rice husk for the removal of Microcystin-LR (MCLR) from water, since it is a relatively underexplored target compared to

dyes and heavy metals. Also, the use of rice husk provides a sustainable, low-cost, circular-economy approach to producing adsorbents. Modification with MgO increased surface basicity and the number of active adsorption sites, resulting in improved removal efficiency compared to the unmodified biochar. Moreover, the work is environmentally relevant, given the increasing incidence of cyanobacterial harmful algal blooms and the need for efficient, locally available solutions for cyanotoxin-contaminated water. However, while the material demonstrated strong adsorption performance for Microcystin-LR, its stability and adsorption efficiency over multiple cycles remain untested. Future studies are recommended to investigate the regeneration potential and long-term usability of the adsorbent to better assess its practical applicability in water treatment systems. Further studies on the adsorption isotherm and kinetics are also recommended to better elucidate the adsorption mechanism and equilibrium behavior.

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#### Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work, the authors used ChatGPT 4o to correct typos and grammar errors. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the content of the published article.

#### Authors' Contributions

Louriejean L. Alfar: Writing – original draft, visualization, methodology, conceptualization. Allen Rhay B. Bayantong: Writing – review & editing, project administration, investigation. Chiena L. Palconite: Writing – review & editing, project administration, investigation. All authors read and approved the final manuscript.

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#### Data availability

The datasets generated and/or analyzed during the current study are available from the corresponding author upon reasonable request.

#### Declarations

##### Competing Interests

The authors declare that they have no competing interests.

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